APPENDIX A
RELATED WORKS

OTHER reservoir simulators include:

- BOAST [1], [2]: Black Oil Applied Simulation Tool is a free simulator from the U.S Department of Energy. It uses IMPES (finite difference, implicit pressure, explicit saturation). The last release was in 1998.
- MRST [3]: Matlab Reservoir Simulation Toolbox is developed by SINTEF Applied Mathematics, which also contributed to OPM. It also includes third-party modules from developers from many different research institutes.
- ECLIPSE [4]: Originally developed by ECL (Exploration Consultants Limited), now owned and developed by Schlumberger. Eclipse is an industry reference simulator. The in- and output files of OPM are compatible with Eclipse software.
- INTERSECT [5]: Also from Schlumberger, it has similar features as Eclipse. Intersect has more support for massively parallel execution, with a GPU-accelerated linear solver and faster linearization of equations.
- ECHELON [6]: Echelon is the only reservoir simulator that is fully GPU accelerated. It uses a GMRES solver with a CPR preconditioner. A maximum of 12 million active cells can be simulated on a single GPU.
- TeraPOWERS [7]: TeraPOWERS is an in-house simulator by Aramco. It boasts the world's first trillion-cell simulation, performed on 150000 cores of the Shaheen II supercomputer [8].
- GEOSX [9]: An open-source simulator, specifically for modeling carbon storage. It does not appear to have a three-phase black oil simulation. There is a possibility of using CUDA, but it is unclear which parts are accelerated exactly.
- tNavigator [10]: A black oil, compositional, and thermal reservoir simulator. Allows the linear solver part to run on a GPU, as well as some other parts.
- PFLOTRAN [11]: An open source simulator that uses PETSc [12] for domain decomposition to achieve parallelism. Not accelerated on GPU but is able to scale easily on CPUs.

Numerous libraries that support blocked sparse linear algebra on GPU exist, including cusparse [13], magma [14] [15], viennacl [16], ginkgo [17], amgcl [18], rocalution [19], and rocsparse [20]. Of these, only cusparse does not support AMG GPUs at all, others might need the HIP [21] module from the ROCm [22] framework. Rocalution actually uses many functions from rocsparse underneath. Table I shows the different libraries and what functions they support. The Chow Patel ILU0 indicates an iterative, highly parallel decomposition [23] and application [24].

Package	ILU0	Chow Patel ILU0	AMG	CPR	bicgstab
cusparse	✓	-	-	-	✓[1]
magma	✓	✓	-	-	✓
viennacl	✓	✓	✓	-	✓
ginkgo	✓	✓	✓	-	✓
amgcl	-	✓	✓	✓	✓
rocalution	✓	-	✓	-	✓
rocsparse	✓	-	-	-	✓[1]

TABLE I: Different sparse linear algebra libraries and their GPU components. [1] bicgstab is not readily available but can be constructed using functions from that library.

APPENDIX B
RESULT TABLES

coupled	DUNE, 1 MPI	388 (65+174+76+69), 2041, 1660, 22346
	DUNE, 4 MPI	139 (25+ 45+24+27), 1989, 1610, 22305
	DUNE, 8 MPI	113 (17+ 31+22+21), 1936, 1596, 21900
	DUNE, 16 MPI	84 (13+ 19+ 6+16), 1909, 1539, 22056
	DUNE, 64 MPI	78 (8+ 16+ 3+14), 1916, 1548, 23318
	rocsparse-0	362 (85+95+98+80), 1946, 1578, 21836
	rocsparse-150	330 (93+47+96+89), 1894, 1528, 25575
separate	DUNE, 1 MPI	438 (77+178+102+77), 1894, 1526, 21920
	DUNE, 4 MPI	125 (21+ 39 + 21+25), 1883, 1519, 22192
	DUNE, 8 MPI	129 (26+ 26 + 29+23), 1894, 1530, 22653
	DUNE, 16 MPI	85 (8 + 17 + 8 +16), 1882, 1519, 23026
	DUNE, 64 MPI	81 (13 + 14+ 3 +15), 1913, 1544, 23815
	rocsparse-0	375 (95 +91+100+83), 1951, 1580, 23049
	rocsparse-150	359 (101+58+103+90), 1953, 1582, 27014

TABLE II: *NORNE* benchmark, EPYC 7763 CPU vs. AMD Instinct MI210, masters of 2023-4-12 10:00

coupled	DUNE, 1 MPI	7015 (992+3832+1344+826), 3933, 3422, 65295
	DUNE, 4 MPI	2272 (330+1283+ 365+244), 3861, 3345, 64765
	DUNE, 8 MPI	1286 (215+ 696+ 156+138), 3915, 3392, 64997
	DUNE, 16 MPI	885 (124+ 424+ 108+ 90), 4058, 3536, 67293
	DUNE, 64 MPI	610 (77 + 267 + 36 + 50), 3799, 3289, 66289
	rocsparse-0	4700 (1238+779+1716+944), 3905, 3387, 65494
	rocsparse-150	5077 (1315+685+2070+982), 4011, 3493, 69223
separate	DUNE, 1 MPI	7019 (981+3847+1334+838), 3936, 3419, 66986
	DUNE, 4 MPI	2153 (331+1162+ 366+240), 3823, 3316, 67790
	DUNE, 8 MPI	1198 (196+ 618+ 160+132), 3865, 3344, 66264
	DUNE, 16 MPI	774 (109+ 392+ 117+ 85), 3678, 3176, 64504
	DUNE, 64 MPI	562 (75 + 212 + 33 + 49), 3799, 3292, 68168
	rocsparse-0	5176 (1470+873+1805+1000), 4532, 3963, 82683
	rocsparse-150	4943 (1359+867+1755+936), 4495, 3940, 85263

TABLE III: *NORNE refined* benchmark, EPYC 7763 CPU vs. AMD Instinct MI210, masters of 2023-4-12 10:00

separate	DUNE, 1 MPI	9150 (2209+3755+2859+294), 2763, 2465, 17419
	DUNE, 4 MPI	2513 (623+1097+ 601 +132), 2802, 2514, 17648
	DUNE, 8 MPI	1390 (333+ 593 + 308 +103), 2834, 2547, 18144
	DUNE, 16 MPI	888 (219+ 359 + 161 + 98), 2802, 2510, 17942
	DUNE, 64 MPI	528 (125 + 215 + 50 + 78), 2675, 2384, 18104
	rocsparse-0	6696 (2738+ 804+2816+298), 2666, 2379, 17621
	rocsparse-150	6156 (2462+ 955+2380+282), 2782, 2488, 17244

TABLE IV: *Big model* benchmark, EPYC 7763 CPU vs. AMD Instinct MI210, masters of 2023-4-12 10:00

	Benchmark:	NORNE		NORNE Refined		Big Model	
Profiled Kernel	Profiled Metric	0 blocks	150 blocks	0 blocks	150 blocks	0 blocks	150 blocks
scale_bsrsv2	Compute(SM)(%)	7	7	18	18	22	22
	SM Busy (%)	15	15	24	24	24	24
	Mem BW (GB/s)	156	159	550	540	960	964
	Mem Busy(%)	15	15	47	46	67	67
	L1 Hit Rate(%)	0	0	0	0	0	0
	L2 Hit Rate(%)	56	55	52	52	59	59
	Occupancy (%)	49	49	77	76	81	81
solve_lower	Compute(SM)(%)	23	23	27	34	33	41
	SM Busy (%)	19	30	28	40	34	43
	Mem BW (GB/s)	11	40	40	76	65	104
	Mem Busy(%)	23	18	29	27	30	32
	L1 Hit Rate(%)	1	4	2	5	3	5
	L2 Hit Rate(%)	72	73	75	73	74	72
	Occupancy (%)	81	64	81	75	83	82
solve_upper	Compute(SM)(%)	23	23	27	34	36	42
	SM Busy (%)	19	31	28	43	37	45
	Mem BW (GB/s)	11	42	39	77	72	109
	Mem Busy(%)	23	18	30	27	30	32
	L1 Hit Rate(%)	1	4	2	5	3	5
	L2 Hit Rate(%)	72	74	76	73	75	72
	Occupancy (%)	81	66	81	78	83	78
bsrmv	Compute(SM)(%)	44	43	51	51	53	53
	SM Busy (%)	49	49	52	52	53	53
	Mem BW (GB/s)	297	295	364	364	343	343
	Mem Busy(%)	53	50	59	59	59	59
	L1 Hit Rate(%)	22	22	22	22	21	22
	L2 Hit Rate(%)	29	28	40	40	42	43
	Occupancy (%)	56	56	59	59	59	59

TABLE V: OPM Flow Profiles of the GPU cuSparse Linear Solver using the 0 and 150 Jacobi Blocks Preconditioner Settings on the NVIDIA A100 GPU using the NCU Profiler.

	Benchmark:	NORNE		NORNE Refined		Big Model	
Profiled Kernel	Profiled Metric	0 blocks	150 blocks	0 blocks	150 blocks	0 blocks	150 blocks
bsrilu0_2_8	VALUInsts	1688	700	1352	849	1276	1020
	SALUInsts	1323	456	1000	554	936	702
	VALUUtilization (%)	84	72	80	73	80	77
	VALUBusy (%)	16	22	20	34	20	24
	SALUBusy (%)	12	14	14	21	15	16
	MemUnitBusy (%)	64	46	68	70	70	70
	MemUnitStalled (%)	1	3	4	9	7	11
	L2CacheHit (%)	99	97	99	94	98	95
	WriteSize (KB)	20	16	195	199	640	629
	FetchSize (KB)						
bsrsv_lower	VALUInsts	418	249	300	264	274	260
	SALUInsts	404	120	169	122	140	123
	VALUUtilization (%)	65	49	52	49	50	49
	VALUBusy (%)	15	33	34	44	38	45
	SALUBusy (%)	15	16	18	19	19	21
	MemUnitBusy (%)	54	55	71	73	82	81
	MemUnitStalled (%)	1	3	3	6	12	9
	L2CacheHit (%)	98	86	91	75	84	69
	WriteSize (KB)	1	1	12	28	67	81
	FetchSize (KB)	19	16	160	194	559	722
bsrsv_upper	VALUInsts	433	264	309	270	277	264
	SALUInsts	430	148	183	131	143	128
	VALUUtilization (%)	66	52	53	50	51	50
	VALUBusy (%)	14	23	33	42	39	47
	SALUBusy (%)	14	13	19	20	20	22
	MemUnitBusy (%)	56	56	71	71	86	80
	MemUnitStalled (%)	1	4	3	6	14	8
	L2CacheHit (%)	98	88	91	74	83	66
	WriteSize (KB)	1	1	12	28	62	86
	FetchSize (KB)	17	14	156	199	540	725
bsrxmvn_3x3	VALUInsts	154	154	141	141	136	136
	SALUInsts	31	31	30	30	29	29
	VALUUtilization (%)	66	66	72	72	70	70
	VALUBusy (%)	5	5	6	6	7	7
	SALUBusy (%)	1	1	1	1	1	1
	MemUnitBusy (%)	69	69	93	92	98	98
	MemUnitStalled (%)	5	6	12	12	14	14
	L2CacheHit (%)	62	62	62	63	64	64
	WriteSize (KB)	1	1	8	8	25	26
	FetchSize (KB)	30	30	230	230	361	632

TABLE VI: OPM Flow Profiles of the GPU rocSparse Linear Solver using the 0 and 150 Jacobi Blocks Preconditioner Settings on the AMD MI210 using the rocprofiler.

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